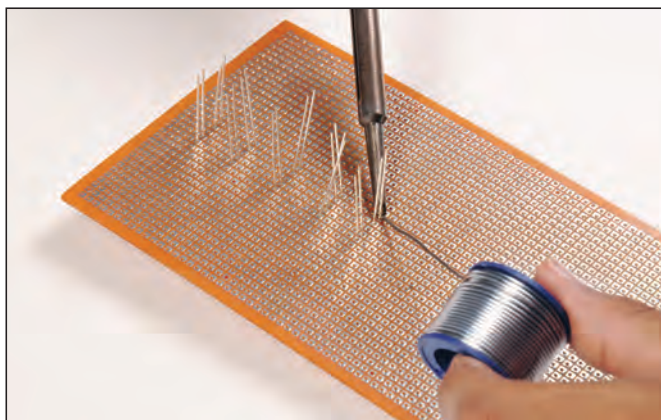


1,60,000

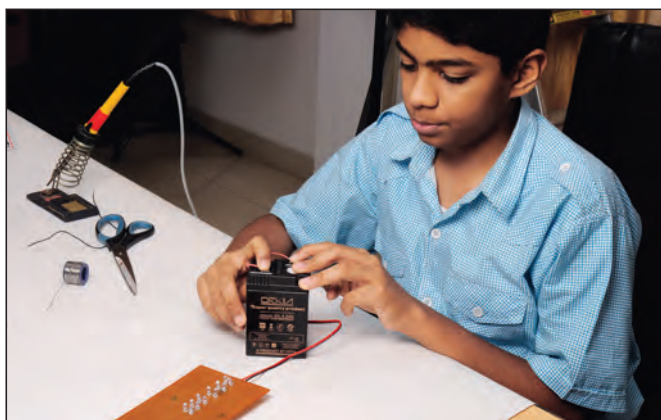
The number of solar photovoltaic panels used in the largest PV plant – the Olmedilla photovoltaic park in Spain

8896 MW

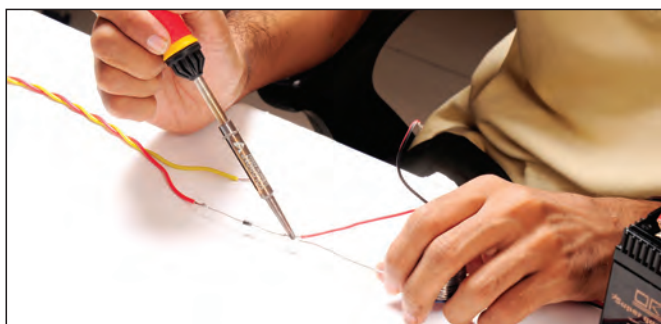
India has the world's fifth largest wind power industry, generating 8896 MW from wind



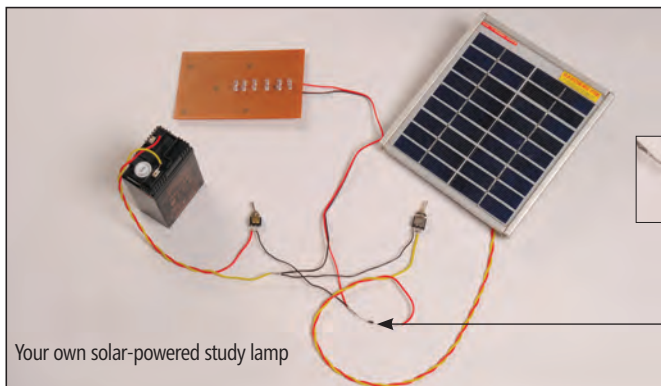
Step 3: Solder the outer leads of the LEDs to ensure they are firmly held to the board



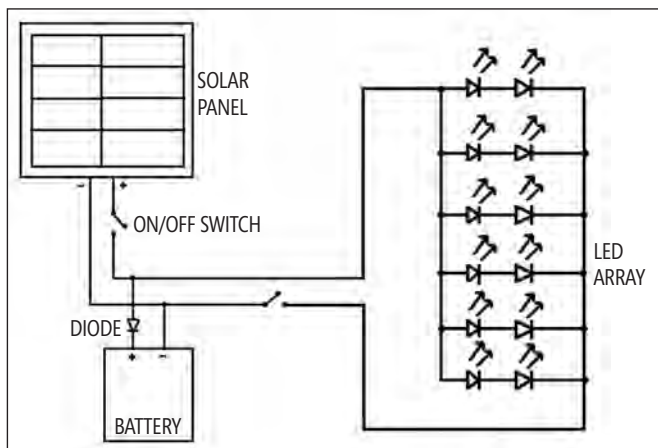
Step 4: Ankit arranged the LEDs with the correct polarity. Here, he's done with the first part of the lamp – the LED array



Step 5: The diode prevents reverse current from flowing towards the solar panel



Your own solar-powered study lamp



The wiring diagram for the solar-powered study lamp

Step 6

Once your LED array is up and running, what you need to do is connect the solar panel, protective diode, and switch.

Step 7

After connecting all the components together, verify the connections with the wiring diagram above.

Step 8

After you have assembled the lamp, you need to put it in an enclosure. You can use a shoebox to house the LED array. The battery can be placed in a plastic jar, with the weight of the battery aiding mechanical stability.

Basic solar-powered system

The basic components required to produce electricity from the sun are solar panels, charge controller, power inverter, and of course, batteries. The number of solar panels you need depends on your power requirement. The charge generated by the solar panels is used to supply the necessary electricity and charge the battery bank. Charge controllers are used to prevent battery overcharging. This prevents battery damage



Don't let your studies be hampered due to lack of a table lamp

and also ensures longer lifespan of the battery. Power inverters form the basis of the solar power system. The basic function of the solar inverter is to convert 12 volts generated from the solar panel to 220 V AC that can be used to drive household appliances.

Considering the fact that sunshine is available for just 8-12 hours a day, it is necessary to be able to store the charge generated by the solar panels. For this, battery banks are used.

We met Sanjay Gupta, proprietor of Solar Spectrum Energy Systems who shared insights into some products marketed by his company. Gupta has agreed to offer these products at some exciting prices specially for Digit readers.